Clinton Enwerem

United States

Academic CV

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Research Interests

Dexterous Manipulation, Robot Learning, Motion Planning, Risk-Sensitive Control, Safety-Critical Control, and State Estimation under Multimodal Uncertainty.

Education

Ph.D., Electrical & Computer Engineering 2021–Fall 2026
University of Maryland, College Park | MD, USA
Advisors: Professor John S. Baras and Professor Calin Belta. **Admitted to Candidacy**: June 2025.
Relevant Coursework: Decision-Making for Robotics, Network Control Systems, Decision Making Under Uncertainty, Random Processes, Advanced Digital Signal Processing, Nonlinear Control Systems, Optimal Control, System Theory, Convex Optimization, Formal Methods for Control & Dynamical Systems, Multimodal Foundational Models.

B.Eng., Electrical Engineering 2014–2018
University of Nigeria, Nsukka | Enugu, Nigeria
Graduated with Highest Honors. *Emphasis*: Control Theory. *Completed coursework in engineering numerical analysis, control systems, electrical machines and drives, computer architecture, electronics, and communication systems*.

Publications

Preprints / Articles In Review

[P5] **Clinton Enwerem**, Saleh Kemal, John S. Baras, and Calin Belta. Belief-Informed Hybrid Control with Almost-Sure Target-Set Convergence, October 2026. Preprint. *2026*

[P4] **Clinton Enwerem**, John S. Baras, and Calin Belta. Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds. arXiv preprint, September 2026. arXiv:2609.01453, project page Code Data and Models. *2026*

[P3] **Clinton Enwerem**, John S. Baras, and Calin Belta. Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality, August 2026. arXiV Link Project Page. *2026*

[P2] **Clinton Enwerem**, John S. Baras, and Calin Belta. FIRMGrasp: A Friction-Informed Risk Margin for Robust Grasp Synthesis, July 2026. arXiV Link. *2026*

[P1] **Clinton Enwerem**, John S. Baras, and Calin Belta. EquiDexFlow: Contact-Grounded SE(3)-Equivariant Dexterous Grasp Generative Flows, June 2026. arXiV Link. *2026*

Refereed Conference Publications

[C6] **Clinton Enwerem**, John S. Baras, and Calin Belta. Risk-Constrained Belief-Space Optimization for Safe Control under Latent Uncertainty, December 2026. *To appear in the Proceedings of the 65th IEEE Conference on Decision and Control (CDC).* arXiV Link. *2026*

[C5] **Clinton Enwerem**, Shreya Kalyanaraman, John S. Baras, and Calin Belta. Variational Neural Belief Parameterizations for Robust Dexterous Grasping under Multimodal Uncertainty, 2026. *Proceedings of the 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).* arXiV Link. *2026*

[C4] **Clinton Enwerem**, Aniruddh G. Puranic, John S. Baras, and Calin Belta. Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression. *2025 IEEE 64th Conference on Decision and Control (CDC)*, pages 4890–4895, December 2025. arXiV Link. *2025*

[C3] **Clinton Enwerem**, Erfaun Noorani, John S. Baras, and Brian M. Sadler. Robust Stochastic Shortest-Path Planning via Risk-Sensitive Incremental Sampling. *2024 IEEE 63rd Conference on Decision and Control (CDC)*, pages 1087–1094, December 2024. arXiV Link. *2024*

[C2] **Clinton Enwerem** and John S. Baras. Safe Collective Control under Noisy Inputs and Competing Constraints via Non-Smooth Barrier Functions. *2024 European Control Conference (ECC)*, pages 3762–3768, June 2024. *2024*

[C1] **Clinton Enwerem** and John S. Baras. Consensus-Based Leader-Follower Formation Tracking for Control-Affine Nonlinear Multiagent Systems. *9th International Conference on Control, Decision and Information Technologies (CoDIT)*, July 2023. *2023*

Refereed Journal Articles

[J1] **Clinton Enwerem** and John S. Baras. Formation Tracking for a Class of Uncertain Multiagent Systems: A Distributed Kalman Filtering Approach. *IEEE Control Systems Letters*, volume 8, 2024. *2024*

Research Experience

Institute for Systems Research, University of Maryland 8/2021–Present
Graduate Research Assistant *College Park, MD*

Research Topics: *Dexterous Manipulation, Robot Learning, Motion Planning, State Estimation, and Feedback Control*

- Developed risk-sensitive distributional RL (**CDC 2025**): **448 goals** versus **329 and 239** across **100 MuJoCo reach–avoid episodes**, with similar constraint-violation costs.
- Trained PPO with grasp-distance-field rewards in **1,024 parallel Isaac Lab environments**; a Unitree G1 with a dexterous hand reached the pregrasp pose in **64/64** evaluation episodes versus **0/64** with palm-distance rewards.
- Fine-tuned OpenVLA-7B with QLoRA for recovery supervision in simulated bimanual packing; each of three training seeds achieved 136/200 task completions versus 129/200 without supervision, recovering seven failures without disrupting nominal successes.
- Built depth-to-trajectory planning with RANSAC/DBSCAN segmentation, point-to-plane ICP registration, and constrained CasADi/IPOPT optimization; verified kinematics, joint limits, and path constraints in MuJoCo.
- Develop grasp-synthesis, state-estimation, motion-planning, and feedback-control methods for dexterous manipulation under friction, pose, and contact uncertainty.
- Derived FIRMGrasp, a CVaR wrench-space margin for grasp selection under friction uncertainty, and implemented Grasp Distance Field control for feedback execution in simulated arm–hand systems.
- Built ParcelStow, an Isaac Lab benchmark for assessing imitation-policy robustness to execution-timing variation; generated **970K** expert states (**297** demonstrations) and trained ACT, Diffusion Policy, and DAgger. Controlled timing stress tests exposed precision-insertion failures: at 2× execution speed, ACT achieved **53%** success versus **84%** for the expert.
- Implement Python and C++ software for articulated robot simulation, sampling-based planning, collision checking, constrained optimization, and manipulation control on fixed-base arms and a 21-DOF fixed-pelvis humanoid model.
- Released ROS 2 manipulation software, including realhand_l6_ros2 for tactile dexterous-hand control over SocketCAN and kinematic_planner_ros2 for sampling-based manipulator planning with FCL collision checking.
- Developed VR teleoperation software for an xArm7-RealHand-L6 robotic manipulator with Cartesian velocity control, relative-joint-position hand control, grasp primitives, thumb-abduction control, and hand-state reset; integrated MoveIt Servo with Dimensional OS (DimOS) for collision detection, arm speed modulation, and velocity damping near kinematic boundaries.
- Published first-author work at CoDIT, ECC, CDC, and IEEE L-CSS, with additional papers to appear at IROS 2026 and CDC 2026 and preprints on manipulation and robot learning.

Electrical Engineering Department, University of Nigeria 9/2018–3/2021
Research Assistant *Enugu, Nigeria*

Research Topics: *Robust Control, Observer-Based Compensator Design, Feedback Linearization*

- Designed and implemented robust nonlinear control, active disturbance rejection, feedback linearization, and observer-based compensation for hardware motor control, released as open-source software, ADRC-Motor-Control.
- Co-authored the resulting peer-reviewed journal paper.

EE Department, University of Nigeria 8/2017–10/2017
Undergraduate Research Assistant *Enugu, Nigeria*

Research Topics: *Feedback Control, Time-Delayed Systems, System Identification*

- Designed feedback controllers for first-order-plus-dead-time processes with dead-time compensation and closed-loop system identification, with the implementation and write-up released as foptd-pade.

Professional Experience

Summer Research Assistant Jun.–Aug. 2023
Institute for Systems Research, University of Maryland | College Park, MD

Supervisor: *Professor John S. Baras*

- Formulated chance-constrained control for multi-agent systems with competing safety constraints.
- Derived Boolean compositions of control barrier functions and evaluated the controller in simulation; the resulting work was published at the 2024 European Control Conference.

MATRIX Lab, USM at Southern Maryland Jun.–Aug. 2022
Research Intern *California, MD*

Supervisor: *Dr. Danilo Romero*

- Conducted system-identification experiments for a Crazyflie quadrotor model.
- Implemented a Lagrangian optimal-control method in ROS-compatible Python for formation tracking with 10 Crazyflie quadrotors under localization uncertainty.

Kognitive Robotics 3/2021–8/2021
Robotics Engineer *Lagos, Nigeria*

Kognitive Robotics is a robotics startup developing turnkey robotics education platforms.

- Designed power and sensor–actuator interfaces in Altium; assembled the mobile robot and integrated its sensors and actuators.
- Implemented the ROS software stack and tested hardware–software integration on the physical robot.

Robotics & Artificial Intelligence Nigeria (RAIN) 1/2020–2/2021
Robotics Trainee *Ibadan, Nigeria*

- Built and tested robotics and IoT prototypes spanning mechanical design, sensor fusion, control firmware, and embedded software.
- Developed a ROS-based quadrotor autonomy stack on a Raspberry Pi 3, integrating ORB-SLAM3 visual SLAM, an EKF fusing SLAM poses with IMU data, and a PID flight controller. The onboard controller maintained stable hover during ground-station packet loss.

Teaching Experience

ECE Department, University of Maryland 9/2025–12/2025
Graduate Teaching Assistant *College Park, MD*

Course: *ENEE467: Robotics Project Laboratory* (with Prof. Calin Belta)

- Mentored 50+ undergraduate students in kinematics, 3D perception, motion planning, and ROS 2 control.
- Built reproducible Docker-based labs and testing pipelines.
- Led weekly lab sessions and provided technical feedback on documentation and debugging.

ECE Department, University of Maryland 2/2025–5/2025
Graduate Teaching Fellow *College Park, MD*

Course: *ENEE661: Nonlinear Control Systems* (with Prof. John S. Baras)

- Led problem-solving sessions on nonlinear analysis and control, covering phase-plane and bifurcation analysis, and stability concepts including Lyapunov, input-to-state, L_2, and L_∞ stability.

Technical Skills

Robotics: ROS/ROS2, ros2_control, MoveIt 2, MoveIt Servo, VR teleoperation, DimOS, MuJoCo, Drake, Gazebo, Isaac Sim, Isaac Lab, Pinocchio, RViz, URDF/SRDF/MJCF, SocketCAN; **Motion Planning:** RRT*, Informed RRT*, Sampling-Based Planning, FCL Collision Checking, OMPL; **Control & Safe Learning:** Safe & Constrained Reinforcement Learning, Control Barrier Functions, Safety Filters, Constrained MDPs, Model Predictive Control (MPC), QP-based Control, Belief-Space Planning, Lyapunov Stability; **Robot Learning & Generative Models:** Vision-Language-Action (VLA) Models, OpenVLA-7B, QLoRA, Imitation Learning & Behavior Cloning, ACT (Action Chunking with Transformers), DAgger, Diffusion Policies (DP3), Flow-Matching Generative Models, Reinforcement Learning (PPO); **ML:** PyTorch, TensorFlow; **Perception & State Estimation:** FoundationPose, BundleSDF, SAM 2, SAM 3D, PCL, OpenCV; **Testing & Validation:** mock hardware, virtual CAN, launch_testing, gtest/pytest, CI; **GPU Computing:** MuJoCo Warp, JAX; **Programming:** Python, C++, MATLAB, Bash, LaTeX; **Optimization:** CasADi, CVXPY, qpsolvers, IPOPT, Gurobi, Mosek, Pyomo; **RL Sandboxes:** OpenAI Gym, Safety Gymnasium, PyBullet; **Dev Tools:** git, GitHub, GitLab, Docker; **Robots & EoATs:** Fairino FR3, xArm7, UR3e, UR5, Robotiq Hand-E, Unitree G1 (simulation), LEAP Hand v1, RealHand L6, RealHand A7 (Bimanual Arm-Hand System).

Honors & Awards

2024: IEEE CSS Student Travel and Workshop Support: Conference travel award to attend CDC'24.
2022: '22–'23 Microsoft Diversity in Robotics & Autonomy PhD Fellowship, MRC & Microsoft.
2022: ROSCon Diversity Scholarship: Travel grant to attend ROSCon 2022 in Kyoto, Japan.
2021: Finalist, Engineers' League, Pan-African Robotics Competition, Rwanda.
2021: CIT Dean's Fellowship, Carnegie Mellon University, Africa Campus, Kigali, Rwanda.
2021: Dean's Fellowship, University of Maryland, College Park, MD, United States.
2020: Scholar, Stanford Exposure to Research & Graduate Education, Stanford University, CA, USA.
2020: EducationUSA Opportunity Funds Program Scholarship, U.S. Consulate General, Lagos, Nigeria.
2020: Sole Recipient (Nationwide), Door Foundation Leadlight Scholarship, RAIN.
2016–2018: Agbami Science & Technology Scholarship, Chevron: Merit-based undergraduate scholarship.
2015–2018: MTN Foundation Scholarship: Nationwide merit-based undergraduate scholarship.

Professional Training & Development

Certificate in Robot Development & Automation 1/2020–2/2021
Robotics & Artificial Intelligence Nigeria (RAIN)
Completed graduate-level coursework and projects in robotics, control theory, ML, and IoT.

Open Courseware Summer 2022, 2023, 2025; Fall 2025
Bayesian Statistics (*UCSD, Coursera*), Autonomous Navigation for Flying Robots (*TUM, edX*), Principles of Robot Autonomy I & II (*Stanford*), Planning & Learning in Robotics (*UCSD*), Robot Learning (*USC*).

Professional Service, Outreach, & Mentoring

2023–2026: *Peer Reviewer*: Heliyon, MED'23, ECC'24, ACC'25, L4DC'25, CDC'25, CDC'26, ICRA'27, IEEE Open Journal of Control Systems, Elsevier (Robotics & Autonomous Systems).
2021–Present: *Member*, Black in Robotics (BiR): advancing Black representation in academic and industry robotics.
2021–2022: *Mentor*, EducationUSA & iScholar Initiative: Helped underrepresented STEM graduates secure fully funded admission to Stanford, Purdue, and Penn State.